Bandersnatch VRF-AD Specification

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Abstract

This specification defines three Verifiable Random Function with Additional Data (VRF-AD) schemes – IETF VRF, Thin VRF, and Pedersen VRF – built on a transcript-based Fiat-Shamir transform with support for multiple input/output pairs via delinearization. The IETF VRF extends RFC-9381 [1]; the Thin VRF and Pedersen VRF follow the constructions introduced by BCHSV23 [2], with the Pedersen VRF serving as a building block for anonymized ring signatures as described in VG24 [3]. All schemes are instantiated over the Bandersnatch elliptic curve, constructed over the BLS12-381 scalar field as specified in MSZ21 [4].

1. Preliminaries

1.1. Groups and Fields

- $\mathbf{G}$: Bandersnatch curve cyclic group of prime order r.
- $\mathbf{F}$: Scalar field of prime order r (i.e. $\mathbb{Z}_r$).
- $\mathbf{\Sigma}^k$: Octet strings with length $k \in \mathbb{N}$ ($*$ for arbitrary length).
- $\mathcal{O}$: Identity point of $\mathbf{G}$.

The EC group $\mathbf{G}$ is the prime subgroup of the Bandersnatch elliptic curve, in Twisted Edwards form, with finite field and curve parameters as specified in MSZ21 [4]. For this group, `fLen` = `qLen` = 32 and `cofactor` = 4.

- $G \in \mathbf{G}$: Prime order group generator.

  G_x=18886178867200960497001835917649091219057080094937609519140440539760939937304

  G_y=19188667384257783945677642223292697773471335439753913231509108946878080696678

  - Compressed: `0x664197ccb667315e6064e4ee81ad8c3586d5dcba508b7d150f3e12da9e666c2a`

1.2. Notation

- $x \in \mathbf{F}$: Secret key scalar.
- $Y \in \mathbf{G}$: Public key point defined as $x \cdot G$.
- $i \in \mathbf{\Sigma}^*$: VRF input data.
- $I \in \mathbf{G}$: VRF input point.
- $O \in \mathbf{G}$: VRF output point.
- $o \in \mathbf{\Sigma}^k$: VRF output hash.
- T: Transcript state (section 1.10).

1.3. Encoding

- `enc_point`(P): Encodes a point in compressed form. The y coordinate is serialized in little-endian and the most significant bit of the last octet encodes the sign of x. This gives `ptLen` = `fLen` = 32.
- `enc_scalar`(s): Encodes a scalar into 32 octets in little-endian representation.
- `enc_32`(n): Encode integer n as a 4-byte little-endian octet string.
- `dec_scalar`(buf): Interpret octet string buf as a little-endian integer and reduce modulo the prime field order r.

Point deserialization MUST output "INVALID" if the octet-string does not decode to a point on the prime subgroup $\mathbf{G}$. All point arithmetic MUST be performed in $\mathbf{G}$. Points not in $\mathbf{G}$ MUST be rejected at all entry points; accepting a point on the full curve but outside the prime-order subgroup enables small-subgroup attacks that break the VRF relation.

1.4. Constants

- `suite_id` = `[0x01, 0x01, 0x01, 0x01]` — a 4-byte fixed-width identifier encoding the protocol profile:

Name	Byte	Value	Meaning
version	0	0x01	Protocol profile (v1)
curve	1	0x01	Bandersnatch
hash	2	0x01	SHA-512
h2c	3	0x01	Elligator 2, random oracle

The version byte bundles several tightly-coupled choices: transcript construction (HashTranscript), nonce algorithm (RFC-8032), challenge derivation (transcript

squeeze), point encoding (compressed little-endian), and security level (128-bit). Bump this byte when any of these changes.

- `challenge_len` = 16 bytes (128-bit security).
- `expanded_scalar_len` = $\lceil(\lceil\log_2(r)\rceil + 128)/8\rceil = 48$ bytes.

Domain separation tags used throughout the protocol:

Tag	Value	Usage
Challenge	0x01	Challenge derivation
NonceExpand	0x02	Nonce secret expansion
Nonce	0x03	Nonce derivation
PointToHash	0x04	VRF output hashing
Delinearize	0x05	Delinearization scalars
PedersenBlinding	0x80	Pedersen blinding factor
IetfVrf	0x10	IETF VRF scheme identifier
ThinVrf	0x11	Thin VRF scheme identifier
PedersenVrf	0x12	Pedersen VRF scheme identifier

1.5. Secret Key Generation

Implementations should provide a method for deterministic secret generation from seeds. One RECOMMENDED method is described in Appendix A. However, any secure method that outputs uniformly random scalars in $\mathbf{F}$ is acceptable.

The secret key scalar MUST NOT be zero. A zero secret key maps every VRF input to the identity point, making all outputs trivially predictable and proofs forgeable.

1.6. VRF Input

The VRF input point $I \in \mathbf{G}$ is derived from the input octet-string using the `hash_to_curve` method defined in section 3 of RFC-9380 [5], instantiated with the *Elligator 2* map to curve (section 6.8.2) and `expand_message_xmd` with SHA-512 (section 5.3.1).

This is the random oracle (`_RO_`) construction: the input is hashed to two independent field elements, each is mapped to a curve point via Elligator 2, and the results are added.

$$I \leftarrow \texttt{hash_to_curve}(DST, i)$$

The domain separation tag is:

$$DST = \text{"ECVRF_"} \parallel \texttt{h2c_suite_id} \parallel \texttt{suite_id}$$

where `h2c_suite_id` = `"Bandersnatch_XMD:SHA-512_ELL2_RO_"` is the RFC-9380 suite identifier string determined by the curve, hash, and h2c bytes of `suite_id`.

Verifiers MUST independently compute each I_i from the corresponding input octet-string using the procedure above. Accepting prover-supplied input points without recomputation breaks the VRF security guarantees, and in the case of Thin VRF (section 3), enables trivial forgery.

1.7. VRF Output

The VRF output point is generated from the VRF input point and secret key scalar:

$$O \leftarrow x \cdot I$$

The VRF output hash is a fixed-length octet string derived from the output point using a transcript-based point-to-hash procedure.

Input:

- $O \in \mathbf{G}$: VRF output point.
- $N \in \mathbb{N}$: Desired output length in bytes.

Output:

- $o \in \mathbf{\Sigma}^N$: VRF output hash.

Steps:

1. $T \leftarrow$ `new_transcript`()
2. T.`absorb`(`PointToHash`)
3. T.`absorb`(`enc_point`(O))
4. $o \leftarrow T$.`squeeze`(N)

1.8. Additional Data

An arbitrary length octet-string provided by the user to be signed together with the generated VRF output. This data doesn't influence the produced VRF output. The length of *ad* MUST NOT exceed $2^{32} - 1$ bytes, as the length is encoded via `enc_32` in the transcript (section 1.11).

1.9. VRF-AD

Regardless of the specific scheme, a *Verifiable Random Function with Additional Data (VRF-AD)* can be concisely represented by three primary functions:

- $prove(x, \overline{io}, ad) \mapsto \Pi$

- $verify(Y, \overline{io}, ad, \Pi) \mapsto (\top \mid \bot)$
- $output(O) \mapsto o$

Where $\overline{io}$ is a sequence of (I_i, O_i) input/output pairs.

1.10. Transcript

The transcript provides a Fiat-Shamir transform with an absorb/squeeze interface. Data is absorbed into an internal hash state; output bytes are squeezed from it. After the first squeeze, `absorb` MUST NOT be called.

Abstract interface:

- `new_transcript`(): Create a fresh transcript instance and absorb `suite_id`.
- `absorb`($data$): Feed bytes into the hash state. MUST NOT be called after squeeze.
- `squeeze`(n) $\rightarrow \mathbf{\Sigma}^n$: Produce n output bytes.
- `fork`(): Clone the transcript state.

Concrete construction (`HashTranscript<SHA-512>`):

Initialization: `new_transcript`() creates a fresh SHA-512 state and feeds `suite_id` into it.

Absorb: feeds raw bytes directly into the SHA-512 state. Consecutive absorb calls are equivalent to a single absorb of the concatenated data. This is safe because all protocol fields use fixed-width encoding (`enc_point`: 32 bytes, `enc_scalar`: 32 bytes, `enc_32`: 4 bytes, domain tags: 1 byte) or explicit length prefixing (ad via `enc_32`(`len`(ad)) $\|$ ad), so the byte stream is unambiguous given the inputs agreed upon by both parties.

Squeeze (counter-mode XOF): on the first squeeze call, finalize the SHA-512 state to obtain a 64-byte *seed*. Then produce output blocks:

$$block_i = \text{SHA-512}(seed \parallel \texttt{enc_32}(i)) \quad \text{for } i = 0, 1, 2, \ldots$$

Each block yields 64 bytes. Output is read sequentially across blocks; partial block state is preserved between squeeze calls.

Fork: duplicates the full internal state (including any partial block position if squeezing has begun).

1.11. VRF Transcript

Shared transcript construction used by all VRF-AD schemes. Absorbs input/output pairs, derives delinearization scalars, merges pairs into a single pair, and absorbs additional data.

Input:

- $scheme$: Scheme identifier tag.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: Sequence of input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- T: Transcript state (with ad absorbed).
- $(I_m, O_m) \in \mathbf{G} \times \mathbf{G}$: Merged input/output pair.

Steps:

1. $T \leftarrow \texttt{new_transcript}()$
2. $T.\texttt{absorb}(scheme)$
3. For each (I_i, O_i) in $\overline{io}$: $T.\texttt{absorb}(\texttt{enc_point}(I_i) \parallel \texttt{enc_point}(O_i))$
4. Delinearize:
   - If $n = 0$: $(I_m, O_m) \leftarrow (\mathcal{O}, \mathcal{O})$
   - If $n = 1$: $(I_m, O_m) \leftarrow (I_0, O_0)$
   - If $n \geq 2$: $T' \leftarrow T.\texttt{fork}()$, $T'.\texttt{absorb}(\texttt{Delinearize})$, squeeze n scalars $z_i \leftarrow \texttt{dec_scalar}(T'.\texttt{squeeze}(\texttt{challenge_len}))$, $I_m \leftarrow \sum_i z_i \cdot I_i$, $O_m \leftarrow \sum_i z_i \cdot O_i$
5. $T.\texttt{absorb}(\texttt{enc_32}(\texttt{len}(ad)) \parallel ad)$
6. Return $(T, (I_m, O_m))$

1.12. Nonce Procedure

Deterministic nonce generation inspired by [RFC-8032] section 5.1.6. The transcript carries shared state from `vrf_transcript`, binding the nonce to the I/O pairs and additional data.

Input:

- $sk \in \mathbf{F}$: Secret scalar.
- T: Transcript state (consumed).

Output:

- $k \in \mathbf{F}$: Nonce scalar.

Steps:

1. $T' \leftarrow T.\texttt{fork}()$
2. $T'.\texttt{absorb}(\texttt{NonceExpand} \parallel \texttt{enc_scalar}(sk))$

3. $h \leftarrow T'.\texttt{squeeze}(64)$
4. $T.\texttt{absorb}(\texttt{Nonce} \parallel h[32..64])$
5. $k \leftarrow \texttt{dec_scalar}(T.\texttt{squeeze}(\text{expanded_scalar_len}))$

Note: T is consumed (mutated then squeezed). Callers must pass forks where the transcript is needed afterwards.

1.13. Challenge Procedure

Derives a challenge scalar by absorbing curve points into the transcript and squeezing.

Input:

- $\bar{P} \in \mathbf{G}^m$: Sequence of m points.
- T: Transcript state (consumed).

Output:

- $c \in \mathbf{F}$: Challenge scalar.

Steps:

1. $T.\texttt{absorb}(\texttt{Challenge})$
2. For each P_i in $\bar{P}$: $T.\texttt{absorb}(\texttt{enc_point}(P_i))$
3. $c \leftarrow \texttt{dec_scalar}(T.\texttt{squeeze}(\texttt{challenge_len}))$

2. IETF VRF

Based on IETF RFC-9381 which is extended with a transcript-based Fiat-Shamir transform, support for additional data (ad), and multiple I/O pairs via delinearization. These changes make this scheme incompatible with standard RFC-9381 implementations and test vectors.

2.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- $\pi = (c, s) \in (\mathbf{F}, \mathbf{F})$: Schnorr-like proof.

Steps:

1. $Y \leftarrow x \cdot G$
2. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\texttt{IetfVrf}, \overline{io}, ad)$
3. $k \leftarrow \texttt{nonce}(x, T.\texttt{fork}())$
4. $k_b \leftarrow k \cdot G, \quad k_h \leftarrow k \cdot I_m$
5. $c \leftarrow \texttt{challenge}([Y, k_b, k_h], T)$
6. $s \leftarrow k + c \cdot x$
7. $\pi \leftarrow (c, s)$

The challenge scalar c is serialized as the first `challenge_len` bytes of its little-endian representation, giving a total proof size of `challenge_len` + `fLen` = 48 bytes. On deserialization, c is interpreted as a little-endian integer with no modular reduction (the value is always less than r).

2.2. Verify

Input:

- $Y \in \mathbf{G}$: Public key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi = (c, s) \in (\mathbf{F}, \mathbf{F})$: Schnorr-like proof.

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. Validate Y and all $I_i, O_i \in \mathbf{G} \setminus \{\mathcal{O}\}$, output $\bot$ if any is invalid or the identity.
2. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\texttt{IetfVrf}, \overline{io}, ad)$
3. $U \leftarrow s \cdot G - c \cdot Y$
4. $V \leftarrow s \cdot I_m - c \cdot O_m$
5. $c' \leftarrow \texttt{challenge}([Y, U, V], T)$
6. $\theta \leftarrow \top$ if $c = c'$ else $\bot$

3. Thin VRF

Thin VRF is derived from the PedVRF construction in section 4 of BCHSV23 [2], reduced to EC-VRF form by setting the blinding factor $b = 0$ and using $pk = sk \cdot G$ directly (see remark on page 13 of the paper).

The scheme merges the Schnorr key pair (G, Y) and VRF I/O pairs into a single DLEQ relation via delinearization, then proves it with a Schnorr-like proof (R, s). Storing the nonce commitment R (rather than the challenge c) enables batch verification.

Security: VRF input points MUST be constructed via hash-to-curve. If a prover knows d such that $I = d \cdot G$, they can forge arbitrary outputs for that input, because the delinearization merges the Schnorr and VRF pairs into a single check that collapses when all points are multiples of G.

3.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- $\pi = (R, s) \in (\mathbf{G}, \mathbf{F})$: Thin VRF proof.

Steps:

1. $Y \leftarrow x \cdot G$
2. $\overline{io}' \leftarrow [(G, Y)] \parallel \overline{io}$ (prepend Schnorr pair)
3. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\texttt{ThinVrf}, \overline{io}', ad)$
4. $k \leftarrow \texttt{nonce}(x, T.\texttt{fork}())$
5. $R \leftarrow k \cdot I_m$
6. $c \leftarrow \texttt{challenge}([R], T)$
7. $s \leftarrow k + c \cdot x$
8. $\pi \leftarrow (R, s)$

3.2. Verify

Input:

- $Y \in \mathbf{G}$: Public key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi = (R, s) \in (\mathbf{G}, \mathbf{F})$: Thin VRF proof.

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. Validate Y, R, and all $I_i, O_i \in \mathbf{G} \setminus \{\mathcal{O}\}$, output $\bot$ if any is invalid or the identity.
2. $\overline{io}' \leftarrow [(G, Y)] \parallel \overline{io}$
3. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\texttt{ThinVrf}, \overline{io}', ad)$
4. $c \leftarrow \texttt{challenge}([R], T)$
5. $\theta \leftarrow \top$ if $s \cdot I_m = R + c \cdot O_m$ else $\bot$

4. Pedersen VRF

Pedersen VRF resembles IETF EC-VRF but replaces the public key with a Pedersen commitment to the secret key, which makes this VRF useful in anonymized ring proofs.

The scheme proves that the output has been generated with a secret key associated with a blinded public key (instead of the public key). The blinded public key is a cryptographic commitment to the public key, and it can be unblinded to prove that the output of the VRF corresponds to the public key of the signer.

This specification mostly follows the design proposed by BCHSV23 [2] in section 4 with some details about blinding base point value and challenge generation procedure.

The *blinding base* $B \in \mathbf{G}$ is defined as follows:

$$B_x = 15502292510512467136772963637174542389568776133586142241717400964712787983 12$$

$$B_y = 28442734166467795856797249030329035618871580593056783094884474814923353898473$$

- Compressed: `0xe93da06b869766b158d20b843ec648cc68e0b7ba2f7083acf0f154205d04e23e`

A point with unknown discrete logarithm derived using the `ECVRF_encode_to_curve` function as described in section 1.6 with input the string: `"pedersen-blinding"`.

4.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- $\pi = (\bar{Y}, R, O_k, s, s_b) \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.

- $b \in \mathbf{F}$: Blinding factor.

Steps:

1. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\texttt{PedersenVrf}, \overline{io}, ad)$
2. $b \leftarrow \texttt{blinding}(x, T.\texttt{fork}())$ (see Appendix A.2)
3. $\bar{Y} \leftarrow x \cdot G + b \cdot B$
4. $T_k \leftarrow T.\texttt{fork}(), \quad T_k.\texttt{absorb}(\texttt{enc_scalar}(b)), \quad k \leftarrow \texttt{nonce}(x, T_k)$
5. $T_{kb} \leftarrow T.\texttt{fork}(), \quad T_{kb}.\texttt{absorb}(\texttt{enc_scalar}(x)), \quad k_b \leftarrow \texttt{nonce}(b, T_{kb})$
6. $R \leftarrow k \cdot G + k_b \cdot B$
7. $O_k \leftarrow k \cdot I_m$
8. $c \leftarrow \texttt{challenge}([\bar{Y}, R, O_k], T)$
9. $s \leftarrow k + c \cdot x$
10. $s_b \leftarrow k_b + c \cdot b$
11. $\pi \leftarrow (\bar{Y}, R, O_k, s, s_b)$

The nonce cross-binding is critical: k is bound to b (step 4) and k_b is bound to x (step 5). This prevents secret/blinding recovery from two proofs with the same (secret, input, ad) but different blinding factors.

4.2. Verify

Input:

- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi = (\bar{Y}, R, O_k, s, s_b) \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. Validate $\bar{Y}$, R, O_k, and all $I_i, O_i \in \mathbf{G} \setminus \{\mathcal{O}\}$, output $\bot$ if any is invalid or the identity.
2. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\texttt{PedersenVrf}, \overline{io}, ad)$
3. $c \leftarrow \texttt{challenge}([\bar{Y}, R, O_k], T)$
4. $\theta_0 \leftarrow \top$ if $O_k + c \cdot O_m = s \cdot I_m$ else $\bot$
5. $\theta_1 \leftarrow \top$ if $R + c \cdot \bar{Y} = s \cdot G + s_b \cdot B$ else $\bot$
6. $\theta = \theta_0 \wedge \theta_1$

Note: no public key appears in the verify inputs – verification uses the committed key $\bar{Y}$ from the proof.

4.3. Unblinding

To link a Pedersen VRF proof to a specific public key, the prover reveals the blinding factor b and the verifier checks:

$$\bar{Y} = Y + b \cdot B$$

where $Y \in \mathbf{G} \setminus \{\mathcal{O}\}$ is the claimed public key. The verifier MUST validate Y before accepting the association.

5. Ring VRF

Anonymized ring VRF based on Pedersen VRF (section 4) and Ring Proof as proposed in VG24.

The following configuration specializes VG24 for the concrete scheme:

- **Groups and Fields**:
  - $\mathbb{G}_{\mathbb{K}}$: BLS12-381 prime order subgroup.
  - $\mathbb{F}$: BLS12-381 scalar field.
  - J: Bandersnatch curve defined over $\mathbb{F}$.
- **Polynomial Commitment Scheme**
  - KZG with SRS derived from Zcash powers of tau ceremony.
- **Fiat-Shamir Transform**
  - `ark-transcript`.
  - Begin with empty transcript and "ring-proof" label.
  - Push R to the transcript after instancing.
- Accumulator seed point in Twisted Edwards form:

  S_x=37805570861274048643170021838972902516980894313648523898085159469000338764576

  S_y=14738305321141000190236674389841754997202271418876976886494444739226156422510

  - Compressed: `0x6e5574f9077fb76c885c36196a832dbadd64142d305be5487724967acf9595a0`

  A point with unknown discrete logarithm derived using the `ECVRF_encode_to_curve` function as described in section 1.6 with input the string: `"ring-accumulator"`.

- Padding point in Twisted Edwards form:

  $\square_x$=26287722405578650394504321825321286533153045350760430979437739593351290020913

  $\square_y$=19058981610000167534379068105702216971787064146691007947119244515951752366738

  - Compressed: `0x92ca79e61dd90c1573a8693f199bf6e1e86835cc715cdcf93f5ef222560023aa`

  A point with unknown discrete logarithm derived using the `ECVRF_encode_to_curve` function as described in section 1.6 with input the string: `"ring-padding"`.

- Polynomials domain ($\langle\omega\rangle = \mathbb{D}$) generator:

  ω=49307615728544765012166121802278658070711169839041683575071795236746050763237

- $|\mathbb{D}| = 2048$

5.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- P: Ring prover (encapsulates ring keys and prover index).

Output:

- $\pi_p \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.
- $\pi_r \in ((G_1)^4, (\mathbf{F})^7, G_1, \mathbf{F}, G_1, G_1)$: Ring proof.

Steps:

1. $(\pi_p, b) \leftarrow Pedersen.prove(x, \overline{io}, ad)$
2. $\pi_r \leftarrow Ring.prove(P, b)$

The blinding factor b is derived internally by Pedersen prove (section 4.1, step 2) and forwarded to the ring prover. *Ring.prove* and *Ring.verify* are defined in VG24 [3].

5.2. Verify

Input:

- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi_p \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.

- $\pi_r \in ((G_1)^4, (\mathbf{F})^7, G_1, \mathbf{F}, G_1, G_1)$: Ring proof.
- $V \in (G_1)^3$: Ring verifier (pre-processed commitment).

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. $\theta_0 \leftarrow Pedersen.verify(\overline{io}, ad, \pi_p)$
2. $(\bar{Y}, R, O_k, s, s_b) \leftarrow \pi_p$
3. $\theta_1 \leftarrow Ring.verify(V, \pi_r, \bar{Y})$
4. $\theta \leftarrow \theta_0 \wedge \theta_1$

Appendix A. Recommendations

A.1. Deterministic Secret Key Scalar Generation

The following method derives a secret scalar from a 32-byte seed. It is provided primarily for test vector generation and is not mandated by the specification. Any secure method that produces uniformly distributed scalars in $\mathbf{F}$ is acceptable.

Input:

- $seed \in \mathbf{\Sigma}^{32}$: seed octet-string.

Output:

- $secret \in \mathbf{F}$: secret key scalar.

Steps:

1. $cnt \leftarrow 0$
2. $T \leftarrow$ `new_transcript`()
3. T.`absorb`($seed$)
4. If $cnt > 0$: T.`absorb`($[cnt]$)
5. $sk \leftarrow$ `dec_scalar`($seed$)
6. $secret \leftarrow$ `nonce`(sk, T)
7. If $secret = 0$: increment cnt and go to step 2
8. Return $secret$

The seed is absorbed into the transcript and also passed as a scalar to the `nonce` procedure (section 1.12), ensuring seed entropy flows through both the transcript state and the secret scalar input paths.

A.2. Deterministic Blinding Factor Generation

The following method generates the Pedersen VRF blinding factor deterministically from the secret key and the VRF transcript state, using the nonce function (section 1.12) with a distinct domain separator. It is provided primarily for test vector generation; implementations may use any method that produces a uniformly random scalar in $\mathbf{F}$.

Linkability warning: because b is derived deterministically from (sk, T), two Pedersen VRF proofs with the same secret key, I/O pairs, and additional data will produce the same blinding factor b and therefore the same blinded public key $\bar{Y} = x \cdot G + b \cdot B$. An observer can detect that both proofs originate from the same signer by comparing $\bar{Y}$ values. Applications that require unlinkability across repeated proofs on the same inputs should generate b as a fresh uniformly random scalar rather than using this deterministic method.

Input:

- $sk \in \mathbf{F}$: Secret scalar.
- T: Transcript state (from `vrf_transcript`).

Output:

- $b \in \mathbf{F}$: Blinding factor scalar.

Steps:

1. T.`absorb`(`PedersenBlinding`)
2. $b \leftarrow$ `nonce`(sk, T)

Appendix B. Behavior with Zero I/O Pairs

When $n = 0$, the `vrf_transcript` procedure (section 1.11) sets the merged pair to the identity: $(I_m, O_m) = (\mathcal{O}, \mathcal{O})$. This causes the VRF-specific verification checks to become trivially satisfied, since any scalar multiplication with $\mathcal{O}$ yields $\mathcal{O}$. The Schnorr proof-of-knowledge component, however, remains sound: a valid proof still requires knowledge of the secret key x.

The per-scheme behavior is as follows:

- **IETF VRF**: The verifier computes $U = s \cdot G - c \cdot Y$ (non-trivial) and $V = s \cdot \mathcal{O} - c \cdot \mathcal{O} = \mathcal{O}$ (vacuous). The scheme degenerates to a Schnorr signature on the additional data ad, proving knowledge of x for public key Y.
- **Thin VRF**: The Schnorr pair (G, Y) is always prepended (section 3.1, step 2), so the internal pair count is at least 1 regardless of the user-supplied n. The scheme remains a well-formed Schnorr proof even with zero VRF pairs.

- **Pedersen VRF**: The VRF output check $O_k + c \cdot O_m = s \cdot I_m$ is vacuously satisfied ($\mathcal{O} = \mathcal{O}$), but the commitment check $R + c \cdot \bar{Y} = s \cdot G + s_b \cdot B$ still proves knowledge of the Pedersen commitment opening (x, b).

No VRF output can be derived when $n = 0$, since there are no output points to hash.

Appendix C. Test Vectors

TODO: The test vectors below are stale and must be regenerated after the nonce procedure change (section 1.10) is applied to the reference implementation.

The test vectors in this section were generated using `ark-vrf` libraries revision `bf2d1cf`.

C.1. IETF VRF Test Vectors

Schema:

```
sk (x): Secret key,
pk (Y): Public key,
in (alpha): Input octet-string,
ad: Additional data octet-string,
h (I): VRF input point,
gamma (O): VRF output point,
out (beta): VRF output octet string,
proof_c: Proof 'c' component,
proof_s: Proof 's' component,
```

Vector 1

```
3d6406500d4009fdf2604546093665911e753f2213570a29521fd88bc30ede18,
a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b,
-,
-,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
e7aa5154103450f0a0525a36a441f827296ee489ef30ed8787cff8df1bef223f,
fdeb377a4ffd7f95ebe48e5b43a88d069ce62188e49493500315ad55ee04d744
..2b93c4c91d5475370e9380496f4bc0b838c2483bce4e133c6f18b0adbb9e4722,
439fd9495643314fa623f2581f4b3d7d6037394468084f4ad7d8031479d9d101,
828bedd2ad95380b11f67a05ea0a76f0c3fef2bee9f043f4dffdddde09f55c01,
```

Vector 2

```
8b9063872331dda4c3c282f7d813fb3c13e7339b7dc9635fdc764e32cc57cb15,
5ebfe047f421e1a3e1d9bbb163839812657bbb3e4ffe9856a725b2b405844cf3,
0a,
```

```
-,
8c1d1425374f01d86b23bfeab770c60b58d2eeb9afc5900c8b8a918d09a6086b,
60f32f5ad3e9694b82ccc0a735edb2f940f757ab333cc5f7b0a41158b80f574f,
44f3728bc5ad550aeeb89f8db340b2fceffc946be3e2d8c5d99b47c1fce344b3
..c7fcee223a9b29a64fe4a86a9994784bc165bb0fba03ca0a493f75bee89a0946,
8aa1c755a00a6a25bdecda197ee1b60a01e50787bd10aa976133f4c39179330e,
18c74ffd67e6abc658e2d05ecd3101ddc0c33623823f2395538cf8d39e654f12,
```

Vector 3

```
6db187202f69e627e432296ae1d0f166ae6ac3c1222585b6ceae80ea07670b14,
9d97151298a5339866ddd3539d16696e19e6b68ac731562c807fe63a1ca49506,
-,
0b8c,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
67a348e256d908eb695d15ee0d869efef2bcf9f0fea646e788f967abbc0464dd,
edde0178045133eb03ef4d1ad8b978a56ee80ec4eab8830d6bc6c08003138841
..6657d3c449d9398cc4385d1c8a2bb19bcf61ff086e5a6c477a0302ce270d1abf,
aec4d1cf308cb4cb400190350e69f4fb309255aa738fff5a6ac4ced7538fce03,
54e5d38a76f309ce63ca82465160abd8d75b78805a0b499e60c26436de4a8e01,
```

Vector 4

```
b56cc204f1b6c2323709012cb16c72f3021035ce935fbe69b600a88d842c7407,
dc2de7312c2850a9f6c103289c64fbd76e2ebd2fa8b5734708eb2c76c0fb2d99,
73616d706c65,
-,
672e8c7a8e6d3eca67df38f11d50f3d7dbb26fa8e27565a5424e6f8ac4555dcc,
4d3e0524fc59374f1fdad8e471c695469b45ecf69c1de85c6c1230e888dd4cbe,
36127f8aee7c61048984f0a208bf6d334db9dacbeeeef9ff2d17117e81232832
..1462eb3ef602f5911d77ab11f815eb4154ba95c934e414198ef000a61b4de31a,
b72598f235145a377911caa794ba85820173c4c49b7be3b05d847b2c753e0311,
e8e34ad3131388a88eb7f80bd874f3421c378d4ad45911c4bc16e4cdc17b5716,
```

Vector 5

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
-,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
ad6af59b4b84f18187c694ef374687d13517cb53508ff9dafa37d0c759e9601c,
4c1269d9d161dabd082fc606af979eca7f6c3ab68e78261dc6fb9fbbb98c9704,
```

Vector 6

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
4fa53519bd9d17acae4d1021416557d11b84dd4670b563770c14eb98161eaa08,
0f7f9bee9077427f547e69b919cf8d63823c14b20085fd9516768e0f5e3d3f0e,
```

Vector 7

```
35b877a25c394512292b82bdf8468e98eaf03c79c7fc9d53546dadc5fb75b500,
b0e1f208f9d6e5b310b92014ea7ef3011e649dab038804759f3766e01029d623,
42616e646572736e6174636820766563746f72,
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
6d1dd583bea262323c7dc9e94e57a472e09874e435719010eeafae503c433f16,
09106f062ac07846f3f841f64765527b333575143483855d633f99ccc2e8e306
..e6239ff79a1272cff931e8d0ac6c390328486329118ad40a18b85184da1837ff,
6dbeeab9648505fa6a95de52d611acfbb2febacc58cdc7d0ca45abd8c952ef12,
ce7f4a2354a6c3f97aee6cc60c6aa4c4430b12ed0f0ef304b326c776618d7609,
```

C.2. Pedersen VRF Test Vectors

Schema:

```
sk (x): Secret key,
pk (Y): Public key,
in (alpha): Input octet-string,
ad: Additional data octet-string,
h (I): VRF input point,
gamma (O): VRF output point,
out (beta): VRF output octet string,
blinding: Blinding factor,
proof_pk_com (Y^-): Public key commitment,
proof_r: Proof 'R' component,
proof_ok: Proof 'O_k' component,
proof_s: Proof 's' component,
proof_sb: Proof 's_b' component
```

Vector 1

```
3d6406500d4009fdf2604546093665911e753f2213570a29521fd88bc30ede18,
```

a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b,
-,
-,
-,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
e7aa5154103450f0a0525a36a441f827296ee489ef30ed8787cff8df1bef223f,
fdeb377a4ffd7f95ebe48e5b43a88d069ce62188e49493500315ad55ee04d744
..2b93c4c91d5475370e9380496f4bc0b838c2483bce4e133c6f18b0adbb9e4722,
01371ac62e04d1faaadbebaa686aaf122143e2cda23aacbaa4796d206779a501,
3b21abd58807bb6d93797001adaacd7113ec320dcf32d1226494e18a57931fc4,
8123054bfdb6918e0aa25c3337e6509eea262282fd26853bf7cd6db234583f5e,
ac57ce6a53a887fc59b6aa73d8ff0e718b49bd9407a627ae0e9b9e7c5d0d175b,
0d379b65fb1e6b2adcbf80618c08e31fd526f06c2defa159158f5de146104c0f,
e2ca83136143e0cac3f7ee863edd3879ed753b995b1ff8d58305d3b1f323630b,

Vector 2

8b9063872331dda4c3c282f7d813fb3c13e7339b7dc9635fdc764e32cc57cb15,
5ebfe047f421e1a3e1d9bbb163839812657bbb3e4ffe9856a725b2b405844cf3,
0a,
-,
-,
8c1d1425374f01d86b23bfeab770c60b58d2eeb9afc5900c8b8a918d09a6086b,
60f32f5ad3e9694b82ccc0a735edb2f940f757ab333cc5f7b0a41158b80f574f,
44f3728bc5ad550aeeb89f8db340b2fceffc946be3e2d8c5d99b47c1fce344b3
..c7fcee223a9b29a64fe4a86a9994784bc165bb0fba03ca0a493f75bee89a0946,
99ff52abf49d67c4303ac4a8a00984d04c06388f5f836ebd37031f0e76245815,
c1322e7a65b83996c25e37a84e36598333b0d417619242c0cb3d9d972edde848,
7a4363e0bf9cd18317287d681ab05704982b0088ce373f696dbdf3909a902b36,
fc8770c209212640742d53e2f40e5c30fffae574f90fdc670ff11a1127586c03,
93f7c9d73eec05e500b758f645a2967e62b2206e57eff5f9b99bfc71812e620d,
c864de36e0b428f6fb4ef470f94ec9601716cb26ad96f3359e4a1ec110794a0b,

Vector 3

6db187202f69e627e432296ae1d0f166ae6ac3c1222585b6ceae80ea07670b14,
9d97151298a5339866ddd3539d16696e19e6b68ac731562c807fe63a1ca49506,
-,
-,
0b8c,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
67a348e256d908eb695d15ee0d869efef2bcf9f0fea646e788f967abbc0464dd,
edde0178045133eb03ef4d1ad8b978a56ee80ec4eab8830d6bc6c08003138841
..6657d3c449d9398cc4385d1c8a2bb19bcf61ff086e5a6c477a0302ce270d1abf,
e22ec3e4a2a4132237eb8a62bcc5ed864593cfde08e53b1632ecd3245761c808,
54c04f259f9e40ee086031d29960b12b6b6407e9de14985001c7265587941831,

```
9200b650a0c20b0ef73ccd7651ffc7af154e5e02879dc8666025c245aa547f01,
35f8dc0f744d1850513c46b6b4640716cbb4643da26cfe67f8c701486e0b4cae,
5faa89369589174f4202d6e53e8b4ef10a49b2ad8face60d7cb28bfc8f43bf0e,
017093ff8d22ba2f3852141365a1452fbb5ab8cf6f20cb04555e3163f8d88f13,
```

Vector 4

```
b56cc204f1b6c2323709012cb16c72f3021035ce935fbe69b600a88d842c7407,
dc2de7312c2850a9f6c103289c64fbd76e2ebd2fa8b5734708eb2c76c0fb2d99,
73616d706c65,
-,
-,
672e8c7a8e6d3eca67df38f11d50f3d7dbb26fa8e27565a5424e6f8ac4555dcc,
4d3e0524fc59374f1fdad8e471c695469b45ecf69c1de85c6c1230e888dd4cbe,
36127f8aee7c61048984f0a208bf6d334db9dacbeeeef9ff2d17117e81232832
..1462eb3ef602f5911d77ab11f815eb4154ba95c934e414198ef000a61b4de31a,
755610da34cc224fbe60ce5e42add2ea6b272ef466aef18c13497363116d1c03,
d26274e014ebfc19a9c1a951193858b972eae3360ed35635e89f1f9dbe432be5,
26202144ba4c4cb7ecde831c9e9662bec519493b29a098dd5803a8b4d261fc12,
b9fa51c75d278d95f2ccace9609b28ec137b244c8b7d1523b16ed07c8e24b8e4,
e42423127a2ca12d4f199287c8fa07784eacf9fc9b86a6bd56ee364cc352c009,
5371d6f9c76b560b4e42b9154a395bed60924d8de31284e926d06af382f5ad1b,
```

Vector 5

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
-,
-,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
fb0123dd6317dbd379afccded247f75b3c1c2e32b86eaa9d6c9d0eb5bef07919,
a91807f0ee57d2344a8942808bf35c65b5bd4fde16752a98f3e3dc67be8c103d,
dbc69ea4dd299deec2f29845e98a17700f07842cf2f6c5e9d88f388fa3a2831c,
311f94e886825c80a30fd44535be37218501bd072afcbc1298f8fba6c3e3c96d,
15f1562046078a7c0d152ef1b56bf3078c763089bf08790f10ff1cf3b9a5030b,
f962cb2598032cf5b0a21b4c253514d75c91ace15acde9a3f716e40f70f06804,
```

Vector 6

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
-,
```

```
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
0752c5b639dffedf9a66ac111a765d3e9c4cfac9c8b26cc5af6d524967afdf0a,
d03caebf8577c1d2ed30a09708683195f11883411dc170e3ea9f09a2cbf86bab,
8b16f0abb2873d6d56199280aeee9e02ce0274a9ca06a3194d6a72c25516ace8,
311f94e886825c80a30fd44535be37218501bd072afcbc1298f8fba6c3e3c96d,
9671cdae8b4cdeea640c24993ccf7e571fcfb3344d81d3cc6f36d03496777c1c,
624e25cd6eccec59b09f0893ef9eab877b55c757b9e9c81260255145bffd9a0d,
```

Vector 7

```
35b877a25c394512292b82bdf8468e98eaf03c79c7fc9d53546dadc5fb75b500,
b0e1f208f9d6e5b310b92014ea7ef3011e649dab038804759f3766e01029d623,
42616e646572736e6174636820766563746f72,
-,
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
6d1dd583bea262323c7dc9e94e57a472e09874e435719010eeafae503c433f16,
09106f062ac07846f3f841f64765527b333575143483855d633f99ccc2e8e306
..e6239ff79a1272cff931e8d0ac6c390328486329118ad40a18b85184da1837ff,
462ae9ad651e5caf11247b989fecb5f2b1729479c33b9133388d14fa35dbbd0c,
91f1ca92eeaa0b604faf3e4811c12b44991ea33cf582a529a4bc4429a3b6cc5a,
b69946f270c46ccc59557bd40288a0a27607281da1892328fdb9da2dcb6c73cf,
5a02419120b814a5c81d67096aac728ee9bda5ddf9451cf554d871462a04831a,
bd8c0c1e5e04577c8836e45fb64131d1275309fe28e1d4334b230e3aa639da1a,
d93ccbd393ed88c8165b0a01aabe28c56a53b43e527e7927eeadff006dd22114,
```

C.3. Ring VRF Test Vectors

KZG SRS parameters are derived from Zcash BLS12-381 powers of tau ceremony.

The evaluations for the ZK domain items, specifically the evaluations of the last three items in the evaluation domain $\mathbb{D}$, are set to 0 rather than being randomly generated.

Schema:

```
sk (x): Secret key,
pk (Y): Public key,
in (alpha): Input octet-string,
ad: Additional data octet-string,
h (I): VRF input point,
gamma (O): VRF output point,
out (beta): VRF output octet string,
blinding: Blinding factor,
```

```
proof_pk_com (Y^-): Pedersen proof public key commitment,
proof_r: Pedersen proof 'R' component,
proof_ok: Pedersen proof 'O_k' component,
proof_s: Pedersen proof 's' component,
proof_sb: Pedersen proof 's_b' component,
ring_pks: Ring public keys,
ring_pks_com: Ring public keys commitment,
ring_proof: Ring proof
```

Vector 1

```
3d6406500d4009fdf2604546093665911e753f2213570a29521fd88bc30ede18,
a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b,
-,
-,
-,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
e7aa5154103450f0a0525a36a441f827296ee489ef30ed8787cff8df1bef223f,
fdeb377a4ffd7f95ebe48e5b43a88d069ce62188e49493500315ad55ee04d744
..2b93c4c91d5475370e9380496f4bc0b838c2483bce4e133c6f18b0adbb9e4722,
01371ac62e04d1faaadbebaa686aaf122143e2cda23aacbaa4796d206779a501,
3b21abd58807bb6d93797001adaacd7113ec320dcf32d1226494e18a57931fc4,
8123054bfdb6918e0aa25c3337e6509eea262282fd26853bf7cd6db234583f5e,
ac57ce6a53a887fc59b6aa73d8ff0e718b49bd9407a627ae0e9b9e7c5d0d175b,
0d379b65fb1e6b2adcbf80618c08e31fd526f06c2defa159158f5de146104c0f,
e2ca83136143e0cac3f7ee863edd3879ed753b995b1ff8d58305d3b1f323630b,
7b32d917d5aa771d493c47b0e096886827cd056c82dbdba19e60baa8b2c60313
..d3b1bdb321123449c6e89d310bc6b7f654315eb471c84778353ce08b951ad471
..561fdb0dcfb8bd443718b942f82fe717238cbcf8d12b8d22861c8a09a984a3c5
..a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b
..4fd11f89c2a1aaefe856bb1c5d4a1fad73f4de5e41804ca2c17ba26d6e10050c
..86d06ee2c70da6cf2da2a828d8a9d8ef755ad6e580e838359a10accb086ae437
..ad6fdeda0dde0a57c51d3226b87e3795e6474393772da46101fd597fbd456c1b
..3f9dc0c4f67f207974123830c2d66988fb3fb44becbbba5a64143f376edc51d9,
afd34e92148ec643fbb578f0e14a1ca9369d3e96b821fcc811c745c320fe2264
..172545ca9b6b1d8a196734bc864e171484f45ba5b95d9be39f03214b59520af3
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Vector 2

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Vector 3

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Vector 4

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Vector 5

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Vector 6

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Vector 7

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